V2 GPS Tracking System — Rewrite Progress Report

Date: March 5, 2026
Platform: Traccar-Web V2 (React + MUI) rewriting GPS-server.net V1 (PHP + jQuery)
Frontend: React 18, Material UI 5, MapLibre GL, Recharts, Vite
Backend: Java Traccar (gsi-traccar fork) with custom API extensions
Deployment: Docker + Easypanel on Ubuntu 24.04 (119.235.255.112)

TABLE OF CONTENTS

1. Architecture Summary
2. Overall Progress Score
3. Feature-by-Feature Status
4. Backend API Status
5. Frontend Component Inventory
6. Data Storage Gaps
7. Missing Features — Detailed
8. Known Bugs & Issues
9. What Has Been Committed
10. Pending Commits
11. Recommended Next Steps

1. ARCHITECTURE SUMMARY

V1 (traccar-old) — GPS-server.net v3.23

```
Browser → PHP Files → MySQL Database
         ├── login.php (authentication)
         ├── tracking.php (main SPA page)
         ├── cpanel.php (admin panel)
         ├── func/*.php (37 controller files)
         ├── api/*.php (6 API files)
         ├── server/*.php (GPS data insertion)
         └── inc/*.php (HTML dialog templates)
Frontend: jQuery 2.1.4, jqGrid, Leaflet.js, Flot Charts, jQuery UI
Database: MySQL with gs_* prefix tables (50+ tables)
Session: PHP sessions + MD5 password + cookie remember-me
```

V2 (traccar-web) — Traccar-based rewrite

```
Browser (React SPA) → Java Traccar Backend → H2/MySQL Database
├── traccar-web (frontend)
│   ├── src/main/ (main page + 23 dialog components)
│   ├── src/settings/ (settings UI + 30+ sub-components)
│   ├── src/map/ (14 map modules + 7 sub-directories)
│   ├── src/reports/ (11 report pages)
│   ├── src/login/ (7 login components)
│   └── src/other/ (8 utility pages)
└── gsi-traccar (backend)
    ├── 19 standard Traccar API resources
    ├── 13 custom GSI API resources
    └── 3 legacy compatibility APIs
Frontend: React 18, MUI 5, MapLibre GL, Recharts, Vite, tss-react
Backend: Java 17, Traccar 6.x fork, Jersey REST, Hibernate
WebSocket: Real-time device/position/event updates
```

2. OVERALL PROGRESS SCORE

Category	V1 Features	V2 Done	V2 Partial	V2 Missing	Score
Authentication & Login	5	5	0	0	**100%**
Main Page & Navigation	8	8	0	0	**100%**
Real-time Tracking	6	6	0	0	**100%**
Toolbar Dialogs (16)	16	16	0	0	**100%**
Sidebar Tabs	4	4	0	0	**100%**
History & Playback	8	8	0	0	**100%**
Places (Markers/Routes/Zones)	12	12	0	0	**100%**
Events System	6	6	0	0	**100%**

Category	V1 Features	V2 Done	V2 Partial	V2 Missing	Score
Settings Dialog	9	7	2	0	**89%**
Object Settings	10	10	0	0	**100%**
Reports	8	6	2	0	**88%**
Map Features	10	10	0	0	**100%**
Data Persistence	7	3	1	3	**50%**
Admin / CPanel	8	4	2	2	**63%**
Server Infrastructure	6	5	1	0	**92%**
				OVERALL	**~91%**

---

3. FEATURE-BY-FEATURE STATUS

✅ FULLY IMPLEMENTED (V1 Parity Achieved)

#	Feature	V2 Component	Lines	Notes
1	**Login / Authentication**	`LoginPage.jsx`	390	Email/password, TOTP, OpenID, remember-me, password reset
2	**Registration**	`RegisterPage.jsx`	183	User registration with server toggle
3	**Password Reset**	`ResetPasswordPage.jsx`	120	Email-based reset flow
4	**Main Page Layout**	`MainPage.jsx`	847	Navbar + sidebar + map + dialogs, 15+ toolbar icons
5	**Device List (Objects)**	`DeviceList.jsx` + `DeviceRow.jsx`	905	Grouped, virtualized, context menu, status colors
6	**Device Data List**	`DeviceDataList.jsx`	210	18 configurable data items
7	**Device Info Panel**	`DeviceInfoPanel.jsx`	1,723	Data list + Graph + Messages tabs, playback
8	**Dashboard**	`DashboardDialog.jsx`	721	6 cards: Objects/Events/Maintenance/Tasks/Odometer/Mileage
9	**Show Point**	`ShowPointDialog.jsx`	155	Coordinate input → map marker
10	**Address Search**	`AddressSearchDialog.jsx`	155	Nominatim geocoding, result list
11	**Reports**	`ReportsDialog.jsx`	915	2 tabs, template CRUD, report properties, generated viewer
12	**Tasks**	`TasksDialog.jsx`	240	Full CRUD (localStorage)

#	Feature	V2 Component	Lines	Notes
13	**RI Logbook**	`LogbookDialog.jsx`	230	Filter, driver/passenger/trailer categories
14	**DTC (Diagnostic)**	`DTCDialog.jsx`	230	Code lookup, OBD-II dictionary
15	**Maintenance**	`MaintenanceDialog.jsx`	230	Real API CRUD, distance/hours/date types
16	**Expenses**	`ExpensesDialog.jsx`	260	Full CRUD (localStorage), total calculation
17	**Object Control**	`ObjectControlDialog.jsx` + 6 tabs	1,809	GPRS/SMS/Templates/Schedule — all functional
18	**Image Gallery**	`GalleryDialog.jsx`	175	Grid view, fullscreen preview
19	**Chat**	`ChatDialog.jsx`	225	Bubble UI, send via Traccar commands
20	**Follow**	`FollowDialog.jsx`	310	Dedicated map, real-time tracking, info panel
21	**System Info (About)**	`InfoDialog.jsx`	90	Server version, map config
22	**History & Route**	`HistoryTab.jsx`	835	Route/stops/events, snap-to-roads (OSRM), XLSX export
23	**Events List**	`EventsList.jsx`	559	Paginated, CSV export, map integration
24	**Events Drawer**	`EventsDrawer.jsx`	70	Right-side notification panel
25	**Places — Markers**	`MarkersTab.jsx` + `MarkerDialog.jsx`	1,039	Groups, visibility, icon selector, map click
26	**Places — Routes**	`RoutesTab.jsx` + `RouteDialog.jsx`	1,049	Color picker, draw on map, deviation
27	**Places — Zones**	`ZonesTab.jsx` + `ZoneDialog.jsx`	1,035	Polygon draw, area calculation
28	**Place Groups**	`PlaceGroupDialog.jsx` + `PlaceGroupsDialog.jsx`	567	Group management for all place types
29	**Settings — Objects**	`ObjectsTab.jsx` + `EditDeviceDialog.jsx`	724	8 sub-tabs matching V1
30	**Settings — Sensors**	`SensorsTab.jsx` + `EditSensorDialog.jsx`	1,002	17 sensor types, calibration, formula
31	**Settings — Services**	`ServiceTab.jsx` + `ServiceDialog.jsx`	899	Odometer/hours/days intervals
32	**Settings — Groups**	`GroupsTab.jsx` + `EditGroupDialog.jsx`	526	Device-to-group assignment
33	**Settings — Drivers**	`DriversTab.jsx` + `EditDriverDialog.jsx`	775	Photo upload, identity, contact info

#	Feature	V2 Component	Lines	Notes
34	**Settings — Events**	`EventsTab.jsx` + `EventEditDialog.jsx`	1,383	37+ event types, multi-tab editor
35	**Settings — Templates**	`TemplatesTab.jsx` + `TemplateDialog.jsx`	455	Variable list, CRUD
36	**Settings — WhatsApp/SMS**	`SMSTab.jsx`	316	Gateway config, HTTP/App modes
37	**Settings — User Interface**	`UserInterfaceTab.jsx`	963	Complete UI preferences matching V1
38	**Settings — My Account**	`MyAccountTab.jsx`	562	Contact info, password, usage, API tokens
39	**Settings — Sub Accounts**	`SubAccountsTab.jsx` + `EditSubAccountDialog.jsx`	997	Permissions, access control, auto-login URL
40	**Map — Multi-provider**	`useMapStyles.js`	275	OSM, Google, Bing, Mapbox, ArcGIS tiles
41	**Map — Device Markers**	`MapPositions.js`	359	Clustering, rotation, click handlers
42	**Map — Geofences**	`MapGeofence.js`	138	Circles, polygons, linestrings
43	**Map — Route Lines**	`MapRoutePath.js` + arrows/markers	446	Route rendering with start/end markers
44	**Map — Playback**	`MapPlaybackMarker.js`	245	Animated position replay
45	**WebSocket**	`SocketController.jsx`	160	Real-time devices/positions/events
46	**Context Menu**	In `DeviceRow.jsx`	—	History shortcuts, follow, street view, send command, edit
47	**KML Import/Export**	In History/Places export	—	KML/GPX/CSV export support
48	**Share Position**	Settings page + device share	—	Via Traccar share mechanism

⚠️ PARTIALLY IMPLEMENTED

#	Feature	Status	What's Missing
49	**Passengers Tab**	**UI stub only**	Tab exists but uses hardcoded `useState([])`. No API integration. No edit dialog. Tab label is also missing in SettingsDialog so it's unreachable.
50	**Trailers Tab**	**UI stub only**	Same as Passengers — pure shell with no functionality. Unreachable via UI.
51	**Report Generation (PDF/XLS)**	**Partial**	ReportsDialog has format selector (HTML/PDF/XLS) but actual generation uses Traccar's built-in `/api/reports/*/xlsx` which only outputs XLSX. True PDF generation not implemented.

#	Feature	Status	What's Missing
52	**Report Scheduling**	**Partial**	ScheduledPage exists in `/reports/scheduled` route. ReportsDialog has daily/weekly schedule fields. But the email delivery mechanism is not confirmed working.
53	**Nearest Zone/ Marker**	**Partial**	Implemented in `DeviceInfoPanel.jsx` but `DeviceDataList.jsx` has TODO comments — returns `'-'` for these fields.
54	**Billing**	**Partial**	V1 has full billing with PayPal integration. V2 has no billing system at all. This may be intentional (different business model).

❌ NOT IMPLEMENTED

#	Feature	V1 Reference	Priority	Notes
55	**Tasks Backend API**	`fn_tasks.php`, `gs_object_tasks`	**HIGH**	V2 stores tasks in localStorage only — data is per-browser, not synced across devices or users. Backend has no `/api/tasks` endpoint.
56	**Chat Backend API**	`fn_chat.php`, `gs_object_chat`	**HIGH**	V2 stores chat in localStorage only — no server persistence, no device-to-server messaging. Backend has no `/api/chat` endpoint. Sends commands but doesn't receive device replies.
57	**Expenses Backend Integration**	`fn_expenses.php`, `gs_user_expenses`	**HIGH**	Backend `/api/expenses` API **exists** but frontend ignores it, using localStorage instead. Simply needs wiring up.
58	**Report Templates Backend**	`fn_reports.php`, `gs_user_reports`	**MEDIUM**	Templates stored in localStorage (`gps_report_templates`). Should sync to server for multi-device access.
59	**CPanel — Dedicated Admin Panel**	`cpanel.php`, `fn_cpanel.*.php`	**LOW**	V1 has a separate admin SPA. V2 distributes admin features across Settings pages. Functional but not consolidated.
60	**CPanel — Bulk Email**	`fn_cpanel.php → send_email`	**LOW**	V1 admin can bulk-email all users / selected users. V2 has AnnouncementPage but not the same email functionality.
61	**CPanel — Database Management**	`fn_cpanel.server.php`	**LOW**	V1 admin can backup DB, optimize tables, view unused objects. V2 has none of this (handled at infrastructure level).
62	**CPanel — Server Cleanup**	`fn_cleanup.php`	**LOW**	V1 auto-cleanup of expired users/objects, orphaned records. V2 relies on Traccar built-in maintenance.
63	**Demo Mode**	`demo.php`	**LOW**	V1 has a demo mode with predefined credentials. V2 has none.
64	**Tachograph Module**	Config toggle, disabled in V1 too	**NONE**	Was disabled/unfinished in V1 as well. Not needed.

#	Feature	V1 Reference	Priority	Notes
65	**Object Forwarding**	`forward_loc_data` in gs_objects	**LOW**	V1 can forward device data to another IMEI. Not implemented in V2.
66	**Virtual ACC**	`accvirt` in gs_objects	**LOW**	V1 can calculate virtual ignition from power voltage. Not in V2.
67	**RI Logbook Backend**	`fn_rilogbook.php`, `gs_rilogbook_data`	**LOW**	V2 LogbookDialog merges API events with localStorage entries. No dedicated backend table.
68	**Mobile Web Version**	`mobile/` directory	**NONE**	V1 had separate mobile HTML/JS. V2 is responsive by design — not needed.

4. BACKEND API STATUS

Custom APIs Added to gsi-traccar

API Resource	Path	Status	Frontend Uses It?
AuditResource	`/api/audit`	✅ Working	✅ Yes (AuditPage)
ConfigurationResource	`/api/configurations`	✅ Working	❌ No
EventHistoryResource	`/api/events/history`	✅ Working	✅ Yes (EventsList)
ExpenseResource	`/api/expenses`	✅ Working	❌ **No — uses localStorage**
GeofenceGroupResource	`/api/geofence-groups`	✅ Working	✅ Yes (Places tabs)
HealthResource	`/api/health`	✅ Working	❌ (infra only)
ScheduledCommandResource	`/api/command-schedules`	✅ Working	✅ Yes (ScheduleTab)
ServiceResource	`/api/services`	✅ Working	✅ Yes (Dashboard, ServiceTab)
SoundResource	`/api/sounds`	✅ Working	❌ No
SubAccountResource	`/api/subaccounts`	✅ Working	✅ Yes (SubAccountsTab)
TemplateResource	`/api/templates`	✅ Working	⚠️ Partial (read-only for system templates)

API Resource	Path	Status	Frontend Uses It?
UserSmsConfigResource	`/api/user-sms-configs`	✅ Working	✅ Yes (SMSTab)
UserTemplateResource	`/api/user-templates`	✅ Working	✅ Yes (TemplatesTab)
MarkerResource (legacy)	`/api/markers`	✅ Working	✅ Yes (MarkersTab)
RouteResource (legacy)	`/api/routes`	✅ Working	✅ Yes (RoutesTab)
ZoneResource (legacy)	`/api/zones`	✅ Working	✅ Yes (ZonesTab)

Backend APIs NEEDED But NOT Created

Feature	Needed API	Priority	Complexity
Tasks	`/api/tasks`	HIGH	Medium — need Task model, TaskResource, DB table
Chat	`/api/chat`	HIGH	Medium — need Chat model, per-device storage
Report Templates	`/api/report-templates`	MEDIUM	Low — model + simple CRUD
Share Position	`/api/share-positions`	LOW	Medium — token generation, expiry
RI Logbook	`/api/logbook`	LOW	Low — model + CRUD

---

5. FRONTEND COMPONENT INVENTORY

Total File Count by Area

Area	Files	Total Lines	Status
`src/main/` (dialogs + page)	26	~8,900	✅ Complete
`src/main/places/`	15	~4,300	✅ Complete
`src/main/ObjectControlTabs/`	6	~1,800	✅ Complete
`src/settings/` (top-level)	30	~4,500	✅ Complete
`src/settings/object/`	11	~3,700	⚠️ 2 stubs
`src/settings/object/tabs/`	10	~3,970	✅ Complete
`src/settings/events/`	2	~1,380	✅ Complete
`src/settings/templates/`	2	~455	✅ Complete

Area	Files	Total Lines	Status
`src/settings/subaccounts/`	2	~997	✅ Complete
`src/settings/sms/`	1	~316	✅ Complete
`src/settings/userinterface/`	1	~963	✅ Complete
`src/settings/myaccount/`	1	~562	✅ Complete
`src/map/`	14 + 7 dirs	~3,500	✅ Complete
`src/reports/`	14	~3,000	✅ Complete
`src/login/`	7	~1,060	✅ Complete
`src/other/`	8	~1,340	✅ Complete
`src/` (root)	10	~1,500	✅ Complete
TOTAL	**~160+**	**~42,000+**	**~91%**

Largest Components (Top 10)

#	Component	Lines	Purpose
1	DeviceInfoPanel.jsx	1,723	Bottom panel: data list + graph + messages
2	EventEditDialog.jsx	1,139	Full event rule editor (6 tabs)
3	UserInterfaceTab.jsx	963	All UI preferences
4	ReportsDialog.jsx	915	Report templates + properties + viewer
5	MainPage.jsx	847	Main page with all state management
6	HistoryTab.jsx	835	History route viewer + controls
7	EditSubAccountDialog.jsx	802	Sub-account permissions editor
8	DashboardDialog.jsx	721	6-card dashboard with charts
9	CustomTab.jsx	686	Device custom fields + attributes
10	IconSelectorDialog.jsx	671	Device icon picker with categories

---

6. DATA STORAGE GAPS

Critical: Data stored in localStorage (should be in database)

Data	localStorage Key	V1 DB Table	Risk	Fix Effort
Tasks	`gps_tasks`	`gs_object_tasks`	Data lost on browser clear, not synced across devices/users	Backend: create `/api/tasks` API Frontend: switch from localStorage to API
Chat	`gps_chat`	`gs_object_chat`	No server persistence, no device replies visible to other users	Backend: create `/api/chat` API Frontend: major rewrite for bi-directional
Expenses	`gps_expenses`	`gs_user_expenses`	Backend API already exists (`/api/expenses`) but frontend ignores it	**Easy fix**: change localStorage calls to fetch/POST calls
Report Templates	`gps_report_templates`	`gs_user_reports`	Templates not accessible from other devices	Backend: create `/api/report-templates` or use user attributes

Non-critical: localStorage usage (acceptable)

Data	Key	Purpose	Risk
Panel heights	`deviceInfoPanelHeight` , `deviceListPanelHeight`	UI layout preference	None — cosmetic
Filter state	`filter` , `filterSort` , `filterMap`	Device list filter	None — convenience
Map style	`selectedMapStyle`	Map tile type	None — preference
Report columns	`summaryColumns` , `tripColumns` , etc.	Column visibility	None — preference
Login email	`loginEmail`	Remember email	None — convenience
Language	`language`	UI language	None — also in user attributes
Device visibility	`deviceInfoPanelHeight`	Panel state	None

7. MISSING FEATURES — DETAILED

7.1 Passengers Tab (STUB)

Location: `src/settings/object/PassengersTab.jsx` (177 lines)

Problem: - Uses `const [passengers, setPassengers] = useState([])` — hardcoded empty - No API calls to `/api/drivers` or any passenger endpoint - Add/Edit/Delete buttons are non-functional (calls empty functions) - Tab label missing in SettingsDialog — only 3 tabs shown ("Object", "Group", "Driver") but component expects index 3

V1 Reference: - `fn_settings.passengers.php` — full CRUD with `gs_user_object_passengers` table - Fields: name, assign_id (iButton/RFID), ID number, address, phone, email, description

Fix Required: - Backend: Add Passenger model extending Traccar's existing structure (or use `drivers` with a `type` field) - Frontend: Implement API integration, edit dialog, add tab label in SettingsDialog

7.2 Trailers Tab (STUB)

Location: `src/settings/object/TrailersTab.jsx` (177 lines)

Problem: Identical to PassengersTab — pure UI shell, no backend integration, tab unreachable.

V1 Reference: - `fn_settings.trailers.php` — full CRUD with `gs_user_object_trailers` table - Fields: name, assign_id, model, VIN, plate number, description

Fix Required: Same approach as Passengers — model, API, dialog, tab label fix.

7.3 Tasks Server Persistence

Location: `src/main/TasksDialog.jsx` (240 lines)

Current: All CRUD operations use `localStorage('gps_tasks')`.

Impact: - Tasks created on one browser/device not visible on another - Clearing browser data loses all tasks - No multi-user task visibility (V1 tasks visible to device watchers)

V1 Reference: `fn_tasks.php` — REST commands with `gs_object_tasks` table. Tasks can be assigned to devices, have start/end addresses with coordinates, delivery status tracking.

7.4 Chat Server Persistence

Location: `src/main/ChatDialog.jsx` (225 lines)

Current: - Messages stored in `localStorage('gps_chat')` per device - Sends commands via Traccar API (`/api/commands/send`) — works for outbound - No mechanism to receive device replies (V1 had `gs_object_chat` with side='C' for client messages)

Impact: - One-directional only (server→device possible, device→server not visible in UI) - No multi-user chat history

V1 Reference: `fn_chat.php` — bidirectional chat stored in `gs_object_chat` with `side` field ('S'=server, 'C'=client/device), status tracking (0=pending, 1=delivered, 2=seen).

7.5 Expenses API Integration

Location: `src/main/ExpensesDialog.jsx` (260 lines)

Current: Uses `localStorage('gps_expenses')`.

Backend: `/api/expenses` API **already exists** in gsi-traccar (`ExpenseResource.java`).

Fix: Low effort — replace localStorage read/write with fetch/POST to `/api/expenses`. This is the easiest gap to close.

7.6 SettingsDialog — Hidden Tabs

Location: `src/main/SettingsDialog.jsx` (319 lines)

Problem: The Object → nested tabs only show 3 Tab labels ("Object", "Group", "Driver") but there are 5 NestedTabPanels (index 0-4: Object, Group, Driver, Passenger, Trailer). Passengers (index 3) and Trailers (index 4) are rendered but their Tab labels are missing, making them unreachable.

Fix: Add Tab labels for Passenger and Trailer (requires the tab components to be functional first).

---

8. KNOWN BUGS & ISSUES

#	Issue	Severity	Location	Description
1	**Passengers/ Trailers tabs hidden**	High	`SettingsDialog.jsx`	Only 3 tab labels for 5 tab panels — Passenger & Trailer tabs unreachable
2	**Passengers tab non-functional**	High	`PassengersTab.jsx`	Hardcoded empty state, no API integration
3	**Trailers tab non-functional**	High	`TrailersTab.jsx`	Hardcoded empty state, no API integration
4	**DeviceDataList nearest_zone TODO**	Medium	`DeviceDataList.jsx`	Returns `'-'` for nearest zone (but works in DeviceInfoPanel)
5	**DeviceDataList nearest_marker TODO**	Medium	`DeviceDataList.jsx`	Returns `'-'` for nearest marker (but works in DeviceInfoPanel)
6	**Global window save functions**	Low	`SettingsDialog.jsx`	Uses `window.smsTabSave` / `window.userInterfaceTabSave` pattern — fragile
7	**Deprecated file in tree**	Trivial	`MapRouteDrawer.old.jsx`	Old version kept, should be removed

#	Issue	Severity	Location	Description
8	**Mixed language**	Low	Multiple files	Indonesian ("Tidak digrup", "Sub akun") mixed with English
9	**Sensors dropdown placeholder**	Low	`ReportsDialog.jsx`	Shows "No sensors available" in report properties
10	**Report PDF generation**	Medium	`ReportsDialog.jsx`	Format selector includes PDF but actual output is XLSX only

9. WHAT HAS BEEN COMMITTED

Git Commit History (key commits)

Commit	Description	Files Changed
`7c5043d4`	Dashboard V1 parity — 6 cards with donut/bar charts	DashboardDialog.jsx
`dd2a34f2`	ShowPoint + AddressSearch V1 parity rewrite	ShowPointDialog.jsx, AddressSearchDialog.jsx
`9de16414`	Dialog spacing fix — increased padding	Multiple dialog files
(earlier)	Events system — WebSocket + event rules + notifications	EventsList.jsx, EventEditDialog.jsx, + backend
(earlier)	Places — Markers/Routes/Zones with groups	15 files in places/
(earlier)	Object Control — GPRS/SMS/Templates/Schedule	6 files in ObjectControlTabs/
(earlier)	Settings — all tabs (object, events, templates, WhatsApp, UI, account, subaccounts)	20+ settings files
(earlier)	History + Playback — route/stops/events with OSRM snap	HistoryTab.jsx, DeviceInfoPanel.jsx
(earlier)	Docker deployment — Dockerfile, nginx, Easypanel	Docker files, CI/CD

10. PENDING COMMITS

File	Status	Description
`ReportsDialog.jsx`	✅ Done, **NOT committed**	Complete rewrite: 2 tabs, template CRUD, report properties (V1 parity), generated viewer
Report Properties fix	✅ Done, **NOT committed**	Layout restructured to match V1: side-by-side schedule/time, added sensors/data items/zones fields, Generate button

11. RECOMMENDED NEXT STEPS

Priority 1 — HIGH (Quick Wins)

#	Task	Effort	Impact
1	**Wire Expenses to API** — Replace localStorage with `/api/expenses` calls	2 hours	Fixes data persistence, API already exists
2	**Fix Passengers/Trailers tabs** — Add tab labels in SettingsDialog, implement API integration	4-6 hours	Completes Settings parity
3	**Fix DeviceDataList nearest_zone/marker** — Copy logic from DeviceInfoPanel	1 hour	Removes TODOs, completes data list
4	**Commit Reports changes** — Push ReportsDialog + Report Properties fixes	15 min	Preserves completed work

Priority 2 — HIGH (Requires Backend Work)

#	Task	Effort	Impact
5	**Create Tasks API** — Backend model + resource + frontend integration	1-2 days	Enables multi-device task management
6	**Create Chat API** — Backend model + resource + bidirectional support	2-3 days	Enables true device↔server chat
7	**Report Templates to server** — Backend endpoint or user attributes sync	1 day	Persists templates across devices

Priority 3 — MEDIUM

#	Task	Effort	Impact
8	**Report PDF export** — Add server-side PDF generation or use browser print	1-2 days	Complete report output format support
9	**Report email scheduling** — Verify cron/scheduled email delivery works	1 day	Automated report delivery
10	**Clean up deprecated files** — Remove MapRouteDrawer.old.jsx	5 min	Code hygiene
11	**Language consistency** — Standardize Indonesian/English strings	2-3 hours	UI polish

Priority 4 — LOW (Nice to Have)

#	Task	Effort	Impact
12	**Admin Dashboard** — Consolidated admin panel (V1 CPanel equivalent)	3-5 days	Admin UX improvement
13	**Billing System** — PayPal/payment gateway integration	5-10 days	Monetization feature
14	**Share Position** — Dedicated share link management with expiry	2-3 days	V1 feature parity
15	**Object Forwarding** — Forward location data to another IMEI	1 day	V1 feature parity
16	**Virtual ACC** — Calculate ignition from voltage sensor	1 day	V1 feature parity
17	**RI Logbook API** — Server-side logbook storage	1 day	V1 feature parity

---

APPENDIX A: FILE COUNTS

```
traccar-web/src/
├── main/              26 files    ~8,900 lines
│   ├── places/        15 files    ~4,300 lines
│   └── ObjectControlTabs/  6 files  ~1,800 lines
├── settings/          30 files    ~4,500 lines
│   ├── object/        11 files    ~3,700 lines
│   │   └── tabs/      10 files    ~3,970 lines
│   ├── events/         2 files    ~1,380 lines
│   ├── templates/      2 files      ~455 lines
│   ├── subaccounts/    2 files      ~997 lines
│   ├── sms/            1 file       ~316 lines
│   ├── userinterface/  1 file       ~963 lines
│   └── myaccount/      1 file       ~562 lines
├── map/               14+7 dirs   ~3,500 lines
├── reports/           14 files    ~3,000 lines
├── login/              7 files    ~1,060 lines
├── other/              8 files    ~1,340 lines
├── common/            various     ~2,000 lines
├── store/             various     ~1,000 lines
├── resources/         various       ~500 lines
└── root files         10 files    ~1,500 lines
                       ─────────   ──────────
                       ~160 files  ~42,000+ lines
```

APPENDIX B: V1 vs V2 TECHNOLOGY COMPARISON

Aspect	V1 (PHP)	V2 (React + Traccar)
Frontend Framework	jQuery 2.1.4 + jQuery UI + jqGrid	React 18 + MUI 5 + Recharts
Map Library	Leaflet.js	MapLibre GL (WebGL-based, smoother)
Build System	None (raw PHP/JS)	Vite (ESBuild + Rollup)

Aspect	V1 (PHP)	V2 (React + Traccar)
Backend	PHP (procedural)	Java (Traccar, Spring-like)
Database	MySQL (50+ gs_* tables)	H2/PostgreSQL (Traccar schema)
Auth	PHP sessions + MD5	JWT/Session + bcrypt + TOTP + OpenID
WebSocket	None (polling)	Full WebSocket (real-time)
GPS Protocol Support	Custom PHP parsers	Traccar (200+ protocols)
Deployment	Apache/nginx + PHP-FPM	Docker + nginx reverse proxy
Code Size	~25,000 PHP lines	~42,000 JS/JSX lines (frontend)
Real-time Updates	AJAX polling (5-30s)	WebSocket (instant)
Mobile	Separate mobile/ directory	Responsive design (same codebase)
Admin Panel	Separate CPanel SPA	Integrated Settings pages
API Architecture	GET-based CSV commands	RESTful JSON (OpenAPI)
Event Processing	Server-side PHP	Java Traccar event engine
Charting	Flot (jQuery plugin)	Recharts (React-native)
Table/Grid	jqGrid (jQuery plugin)	Custom MUI Table components

---

This document is auto-generated from codebase analysis. Last updated: March 5, 2026.